Substantial similarity in amygdala neuronal activity during conditioned appetitive and aversive emotional arousal

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The amygdala is important for determining the emotional significance of environmental stimuli. However, the degree to which appetitive and aversive stimuli are processed by the same or different neuronal circuits within the amygdala remains unclear. Here we show that neuronal activity during the expression of classically conditioned appetitive and aversive emotional responses is more similar than expected by chance, despite the different sensory modalities of the eliciting stimuli. We also found that the activity of a large number of cells (> 43%) was correlated with blood pressure, a measure of emotional arousal. Together, our results suggest that a substantial proportion of neuronal circuits within the amygdala can contribute to both appetitive and aversive emotional arousal.

autonomic | electrophysiology | emotion | fear | Pavlovian conditioning

The amygdala plays an important role in emotional processing (1–5). Of particular interest is whether the amygdala encodes one or both of the most commonly described dimensions of emotion (6–12): valence, which ranges from highly positive to highly negative, and arousal, which ranges from calm to excited or agitated.

Consistent with the hypothesis that the amygdala encodes emotional arousal, neuroimaging studies in humans have found activation of the amygdala during the presentation of both positively and negatively valenced arousing stimuli (3, 8–10, 13–17). Several of these studies also found that the degree of amygdala activation correlates with measures of autonomic arousal (13–17). Paralleling these findings, numerous nonhuman animal studies indicate that the amygdala is important for both appetitively and aversively motivated behaviors (18–23) and corresponding changes in autonomic function (18, 21, 22, 24). The amygdala also has a well-established role in the modulation of appetitive and aversive memories (5, 25). Thus, considerable evidence suggests that the amygdala is important for appetitive and aversive arousal. However, the degree to which the same or different populations of cells within the amygdala are involved in appetitive and aversive arousal remains unclear.

One possibility is that there are two distinct types of neuronal circuits within the amygdala that contribute to emotional arousal—one type of circuit dedicated to processing appetitive stimuli and a different type of circuit dedicated to processing aversive stimuli. According to this model (the "different circuits" model), although appetitive and aversive stimuli often elicit similar physiological responses (e.g., increases in blood pressure), dual mechanisms exist within the amygdala for producing the same physiological response because appetitive and aversive stimuli are processed by distinct circuits in the amygdala. This model predicts that there will be an inverse relationship between neuronal activity in the amygdala during appetitive and aversive arousal—cells which have a certain response during one form of arousal should be less likely than other cells to have the same response during the other form of arousal. Another possibility is that the neuronal circuits within the amygdala that contribute to emotional arousal are indifferent to the valence of the stimuli (the "goodness/badness," or appetitive/aversive nature of the stimuli). According to this model (the "same circuits" model), neuronal circuits in the amygdala that contribute to arousal (e.g., by increasing blood pressure) are not dedicated to either appetitive or aversive stimuli but instead can be recruited by both types of stimuli. This model predicts that there will be a positive relationship between neuronal activity in the amygdala during appetitive and aversive arousal—cells which have a certain response during one form of arousal (appetitive or aversive) should be more likely than other cells to have the same response during the other form of arousal (aversive or appetitive, respectively).

Here, we tested these predictions by using an experimental design which allowed us to quantify the degree of similarity between patterns of neuronal activity during conditioned appetitive and aversive stimuli. Rather than use two similar conditioned stimuli (i.e., stimuli from the same sensory modality) and look for differences in neuronal activity after pairing one with a rewarding outcome and the other with an aversive outcome (26–28), we used stimuli from different sensory modalities and looked for similarities in neuronal activity after pairing one with a rewarding outcome and the other with an aversive outcome. Hence, any observed similarity in neuronal activity cannot be ascribed to encoding of similar sensory features (Fig. 1*A*). Importantly, our experimental design is biased toward supporting the different circuits model, because the sensory features of the conditioned stimuli and outcomes they predict are different. One would expect cells to have different responses to the appetitive and aversive conditioned stimuli simply because they receive different sensory inputs (28, 29). Finding that cells have similar responses during the appetitive and aversive conditioned stimuli despite this bias, on the other hand, would be strong support for the same circuits model. The same circuits model also predicts that a substantial proportion of cells in the amygdala, and especially cells with similar changes in activity during conditioned appetitive and aversive stimuli, will be correlated with measures of arousal. We tested this prediction by simultaneously measuring single-cell activity and blood pressure—a component of autonomic arousal that is sensitive to lesions of the amygdala in both appetitive (24) and aversive (21, 22) settings—and looking for systematic relations between the two during appetitive and aversive conditioning sessions.

Results

Sucrose- and Shock-Conditioned Stimuli Induce Appetitive and Aversive Emotional Arousal. Rats ($n = 10$) were first trained to discriminate between two tones of different frequency during appetitive

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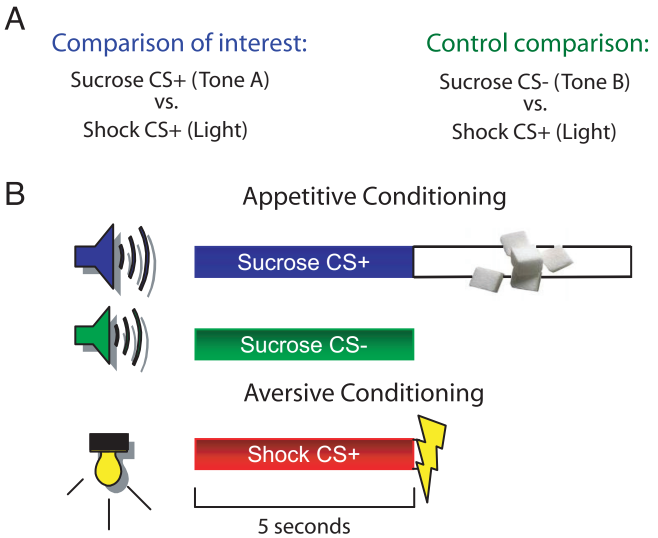

Fig. 1. Experimental design and behavioral training. (*A*) Experimental design. (*B*) Behavioral training. All conditioned stimuli lasted 5 seconds. Sucrose (0.1 mL of a 5% solution, 5 s) was delivered into the reward port immediately after the termination of the sucrose CS+. Footshock (0.4 mA, 0.5 s) was administered immediately after the termination of the shock CS+.

Pavlovian conditioning. One tone [sucrose conditioned stimulus (CS+)] was paired with sucrose delivery in a reward port, whereas the other tone (sucrose CS−) was explicitly unpaired with sucrose (Fig. 1*B*). After 12 sessions, rats approached and remained in the port more often during the sucrose CS+ than the sucrose CS− (*t* test, $P < 0.0001$) (Fig. 2*A*), and blood pressure was elevated during the sucrose CS+ compared with the sucrose CS− (*t* test, $P < 0.01$) (Fig. 2*B* and Fig. S1). The larger blood-pressure response to the sucrose CS+ could not be explained by movement during the sucrose CS+ or its sensory properties (Fig. S2 and *Data and Discussion* in the *SI Text*). Thus, the rats discriminated between the sucrose CS+ and the sucrose CS− after training, indicating that changes in blood pressure during the CS+ reflect motivational, attentional, and/or emotional state.

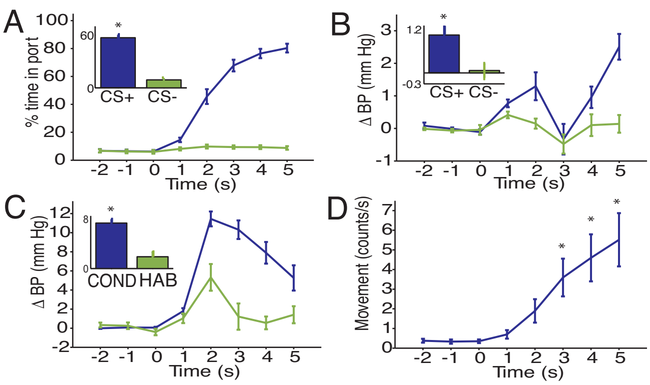

Fig. 2. Behavior and blood pressure during Pavlovian conditioning. (*A*–*D*) Large frames depict mean behavior and blood-pressure responses during the 3 seconds before (bins −2, −1, 0) and 5 seconds after (bins 1–5) conditioned stimulus onset. Insets show the overall mean response during the conditioned stimulus (bins 1–5 together). (*A*) Rats approached and remained in the port longer during the sucrose CS+ than the sucrose CS−. (*B*) Larger increases in blood pressure during the sucrose CS+ than the sucrose CS−. (*C*) Larger increases in blood pressure during aversive conditioning than during habituation. (*D*) Increase in movement during the end of the shock CS+ during aversive conditioning. Bins 3–5 are significantly different from bin 0. (n = 10 rats). * $P < 0.01$.

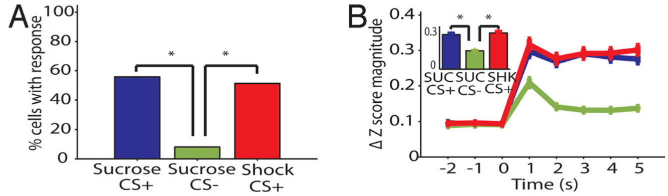

Fig. 3. Neuronal activity during the conditioned stimuli. (*A*) Percentage of cells with statistically significant changes in activity during the conditioned stimuli (n = 518 cells). There were more cells with responses during the sucrose CS+ and shock CS+ than the sucrose CS−. (*B*) Mean magnitude change in neuronal activity during the conditioned stimuli. Mean activity of the population changed more during the sucrose CS+ and shock CS+ than during the sucrose CS− (n = 518 cells). Frames and insets as in Fig. 2. * $P < 0.0001$.

After the discrimination acquisition, rats continued daily conditioning sessions in which the appetitive conditioning session was followed immediately by an aversive conditioning session, forming joint appetitive/aversive conditioning sessions, which are the source of the data presented here. During the first appetitive/aversive conditioning session only, rats were given 10 presentations of the light shock CS+ without footshock ("Habituation" trials) to measure preconditioning changes produced by the shock CS+. The habituation trials were followed by pairings of the shock CS+ with footshock. Consistent with previous reports (21, 22, 30), aversive conditioning increased blood pressure during the shock CS+ relative to habituation trials (*t* test, $P < 0.001$) (Fig. 2*C* and Fig. S1) and produced species-specific defense reactions during the CS (31), i.e., escape/fleeing during the conditioning sessions (mean ± SEM: pre-CS, 0.4 ± 0.1 counts/s; shock CS+, 3.3 ± 0.8 counts/s, *t* test, $P < 0.01$) (Fig. 2*D*), and freezing during a test session without footshock (conducted once per rat at the end of the entire experiment; percentage of time freezing, mean ± SEM: pre-CS, 0 ± 0; shock CS+, 55 ± 12, one-sample *t* test, $P < 0.001$). The changes in behavior and blood pressure we observed during the sucrose CS+ and shock CS+ indicate that these stimuli elicited appetitive and aversive emotional arousal, respectively.

Neuronal Activity Reflects the Emotional Significance of the Conditioned Stimuli. We recorded the activity of 518 cells in the amygdala (basal, lateral, basomedial, and central nuclei) during a total of 106 joint appetitive-aversive conditioning sessions in the 10 rats whose behavior is described above (see Fig. S3 and S4 for cell locations). Most cells had low baseline firing rates, suggesting that we sampled predominantly from putative projection cells (32–34) (Fig. S5).

Previous single-cell recording studies reported robust changes in neuronal activity in the amygdala to auditory or visual stimuli paired with positive or negative reinforcement, but not to stimuli presented repeatedly without consequence (26, 35–39), consistent with a role for the amygdala in assessing the motivational or emotional significance of stimuli. Similarly, we found more cells with significant changes in activity during the sucrose CS+ and shock CS+ than the sucrose CS− (sucrose CS+ vs. sucrose CS−, $\chi^2 = 221.7, P < 0.0001$; shock CS+ vs. sucrose CS−, $\chi^2 = 209.0, P < 0.0001$; sucrose CS+ vs. shock CS+, $\chi^2 = 0.2, P = 0.66$) (Fig. 3*A*). Additionally, the magnitude of the average normalized response across the entire population of recorded cells was larger in response to the sucrose CS+ and shock CS+ than the sucrose CS− (ANOVA, main effect of CS type: $F_{2,1551} = 257.4, P < 0.0001$; main effect of time: $F_{4,2585} = 7.8, P < 0.0001$; CS type x time: $F_{8,5170} = 1.9, P = 0.06$; t tests: sucrose CS+ vs. sucrose CS−, $P < 0.0001$; shock CS+ vs. sucrose CS−, $P < 0.0001$; sucrose CS+ vs. shock CS+, $P = 0.36$) (Fig. 3*B*).

When restricting our analysis to cells with statistically significant responses to the sucrose CS+, we found that they had smaller responses to the sucrose CS− (Fig. S6). This effect was independent

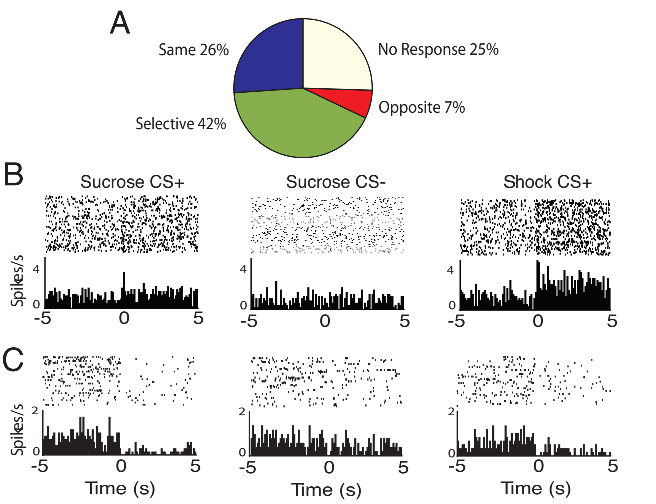

Fig. 4. Categories of neuronal responses to sucrose CS+ and shock CS+. (*A*) Percentage of total population with a given neuronal response type during the sucrose CS+ and shock CS+ (n = 518 cells). (*B* and *C*) Perievent rasters and histograms of a single neuron response to each stimulus. Rasters depict spiking on individual trials; histograms depict mean firing rate across trials. Time on the x axis is relative to stimulus onset ($x = 0$). Bin = 100 ms. (*B*) Example of a same cell with increases in activity during the sucrose CS+ and shock CS+. This cell was located in the BLA. (*C*) Example of a same cell with decreases in activity during the sucrose CS+ and shock CS+. This cell was located in the CeN.

of the sensory properties of the conditioned stimulus and movement (see *Data and Discussion* in the *SI Text* and Fig. S6). The differences in neuronal activity during the sucrose CS+ and sucrose CS− indicate that neuronal activity in the amygdala, like changes in blood pressure, reflects the emotional significance of the conditioned stimuli.

Despite the parallel between neuronal activity and blood pressure, changes in neuronal activity during the sucrose CS+ and shock CS+ were not secondary to changes in blood pressure, because the majority of neuronal response latencies for the sucrose CS+ and shock CS+ were shorter than blood-pressure latencies (Fig. S6). Thus, neuronal responses to the sucrose CS+ and shock CS+ usually preceded blood-pressure responses, consistent with neuronal activity in the amygdala contributing to changes in blood pressure.

Similar Changes in Neuronal Activity During Conditioned Appetitive and Aversive Stimuli. If the same circuits within the amygdala contribute to appetitive and aversive emotional arousal, then neuronal activity should be similar during conditioned appetitive and aversive stimuli. Consistent with prior studies (27, 28), many cells were selective for the different conditioned stimuli (sucrose CS+ selective, 23% of all cells; shock CS+ selective, 19% of all cells) (Fig. 4*A*; see Fig. S7 for sucrose CS− categories) and a smaller group of cells had opposite changes in activity during the sucrose CS+ and shock CS+ (Fig. 4*A*). Notably, however, we also found a large population of cells (26%) which had qualitatively similar changes in activity during the sucrose CS+ and shock CS+ ("same" cells), despite the different modalities of the stimuli, the different outcomes they predicted, and the different behaviors they elicited (Fig. 4 *A*–*C*). Same cells had, on average, larger responses to the sucrose CS+ than the sucrose CS− and faster responses to the sucrose CS+ and shock CS+ than blood pressure, although these effects were not restricted to same cells (Fig. S8 and *Data and Discussion* in the *SI Text*). Same cells were common in the basolateral (BLA) region (basal, lateral, and basomedial nuclei, 26% of cells) and central nucleus (CeN) (28% of cells) of the amygdala.

Although categorizing cells as same, selective, and opposite is useful for an initial description of the population and for comparing the properties of cells with different degrees of similarity, it does not on its own indicate whether there was a relationship between neuronal activity during the sucrose CS+ and shock CS+. To determine whether there was a relationship between neuronal activity during the sucrose CS+ and shock CS+, we computed the probabilities of cells having a specific response to one CS+ (either increase or decrease in activity) given that they had a specific response to the other CS+. We found that, compared to what one would expect by chance (i.e., if responsiveness to the two conditioned stimuli were completely independent), a cell was more likely to have a significant increase in activity during one CS+ if it had a significant increase in activity during the other CS+ ($\chi^2 = 24.9, P < 0.0001$; Fig. 5*A*). The same was true for decreases in activity – a cell was more likely to have a significant decrease in activity during one CS+ if it had a significant decrease in activity during the other CS+ ($\chi^2 = 19.4$, $P < 0.0001$; Fig. 5*B*). Importantly, the opposite relationships did not exist. Cells with increases or decreases in activity during one CS+ were not more likely to have the opposite change in activity; in fact, cells with decreases in activity during the shock CS+ were less likely to have increases in activity during the sucrose CS+ ($\chi^2 = 8.3, P < 0.01$; Fig. 5*A*), although the converse was not true ($\chi^2 = 2.1, P = 0.15$; Fig. 5*B*). Although relationships between responses to the sucrose CS+ and shock CS+ were seen in both the BLA and CeN, they were only significant in the CeN when considering decreases in activity [BLA: (P(sucrose CS+ increase|shock CS+ increase), $\chi2 = 26.0$, $P < 0.0001$; Fig. 5*C*]. [P(sucrose CS+ increase|shock CS+ decrease), $\chi2 = 5.9, P < 0.05$; Fig. 5*C*.] [P(sucrose CS+ decrease|shock CS+ decrease), $\chi2 = 12.4$, $P < 0.0001$; Fig. 5*D*.] [P(sucrose CS+ decrease|shock CS+ increase), $\chi2 = 6.2$, $P < 0.05$; Fig. 5*D*.] [CeN: P(sucrose CS+ decrease|shock CS+ decrease), $\chi2 = 6.1$, $P < 0.05$; Fig. 5*F*).]

We next tested the prediction that the sucrose CS+ and shock CS+ elicit quantitatively similar changes in neuronal activity. To compare neuronal activity during the sucrose CS+ and shock CS+, we computed for each cell the difference in normalized neuronal activity during the two conditioned stimuli and averaged over all of the cells – the "aligned" condition – and compared it to what one would expect if neuronal activity during the stimuli were independent—the "shuffled" condition, obtained by finding the difference in activity during the two stimuli in arbitrary cell pairs (Fig. 6 and *SI Materials and Methods*). Thus, if neuronal activity during the sucrose CS+ and shock CS+ is similar, the difference score in the aligned condition should be less than the difference score in the shuffled condition, and if neuronal activity during the sucrose CS+ and shock CS+ is different, the difference score in the aligned condition should be greater than that in the shuffled condition. Consistent with the hypothesis that neuronal activity during the sucrose CS+ and shock CS+ is similar, the difference score in the aligned condition was indeed less than the difference score in the shuffled condition (ANOVA, main effect of comparison type: $F_{1,1034} = 27.2$, $P < 0.0001$; main effect of time: $F_{4,2585} = 3.7$, $P < 0.01$; comparison type x time: $F_{4,5170} = .4$, $P = 0.79$; inset, t test, $n = 518$ cells, $P < 0.0001$) (Fig. 7*A* and Fig. S9). These findings held true when we examined the BLA and CeN separately [(BLA: ANOVA, main effect of comparison type: $F_{1,626} = 25.1$, $P < 0.0001$; main effect of time: $F_{4,1565} = 2.8$, $P < 0.05$; comparison type x time: $F_{4,3130} = .2$, $P = 0.92$; inset, t test, $n = 313$ cells, $P < 0.0001$ (Fig. 7*C*); CeN: ANOVA, main effect of comparison type: $F_{1,356} = 5.4$, $P < 0.05$; main effect of time: $F_{4,890} = .5$, $P = 0.75$; comparison type x time: $F_{4,1780} = .1$, $P = 0.97$; inset, t test, $n = 178$ cells, $P < 0.05$ (Fig. 7*E*)]. These effects did not depend on baseline firing rates (see *Data and Discussion* in the *SI Text*).

In contrast to the comparison between the sucrose CS+ and the shock CS+, there was no significant difference between the aligned and shuffled conditions for the comparison between the sucrose CS− and the shock CS+ (all amygdala cells: ANOVA, main effect of comparison type: $F_{1,1034} = 1.7$, $P = 0.19$; main effect of time:

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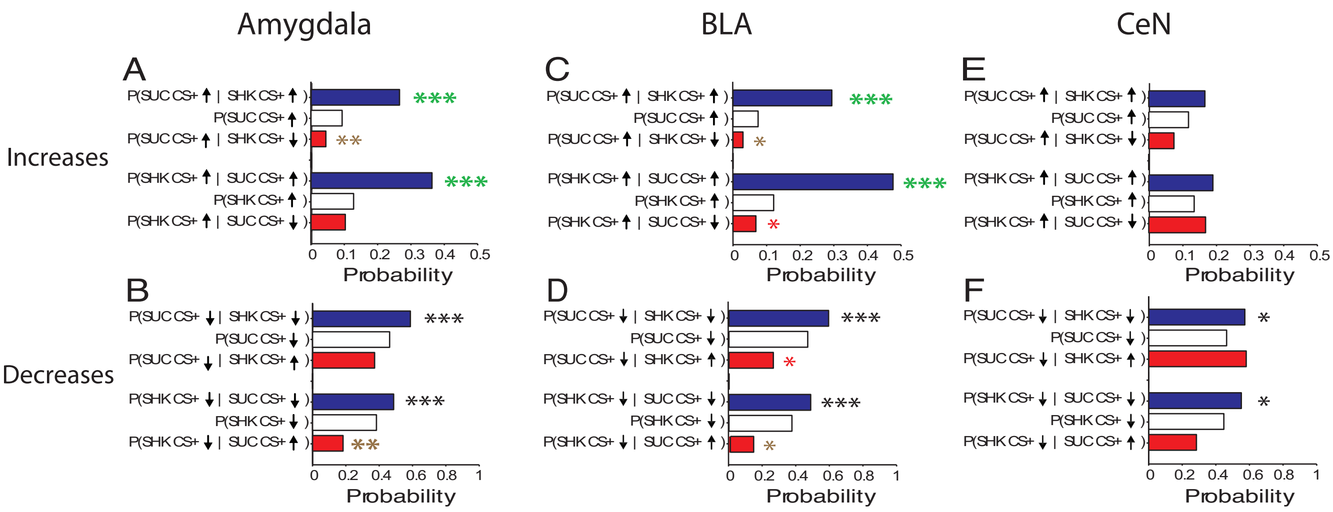

Fig. 5. Neuronal activity during the sucrose CS+ predicts neuronal activity during the shock CS+ and vice versa. (*A* and *B*) Amygdala cells (n = 518 cells). (*C* and *D*) BLA cells (n = 313 cells). (*E* and *F*) CeN cells (n = 178 cells). (*A*, *C*, *E*) Probability of increases in activity. (*B*, *D*, *F*) Probability of decreases in activity. (*A*) Cells with increases in activity during one CS+ were more likely than the rest of the population to have increases in activity during the other CS+. Cells with decreases in activity during the shock CS+ were less likely than the rest of the population to have increases in activity during the sucrose CS+. (*B*) Cells with decreases in activity during one CS+ were more likely than the rest of the population to have decreases in activity during the other CS+. Cells with increases in activity during the sucrose CS+ were less likely to have decreases in activity during the shock CS+. (*C*) Same as *A* except that, additionally, cells with decreases in activity during the sucrose CS+ were less likely than the rest of the population to have increases in activity during the shock CS+. (*D*) Same as *B* except that, additionally, cells with increases in activity during the shock CS+ were less likely than the rest of the population to have decreases in activity during the sucrose CS+. (*F*) Cells with decreases in activity during one CS+ were more likely than the rest of the population to have decreases in activity during the other CS+. Note that half of the statistical tests depicted here are redundant for each dataset (see *SI Materials and Methods*) and denoted as such by the same-color asterisk. Also, note the different scales on the x axis for (*A*, *C*, *E* and *B*, *D*, *F*). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.0001$.

$F_{4,2585} = 4.5, P < 0.01$; comparison type x time: $F_{4,5170} = .7, P = 0.62$; n = 518 cells (Fig. 7*B*); BLA: ANOVA, main effect of comparison type: $F_{1,626} = .89, P = 0.3$; main effect of time: $F_{4,1565} = 4.71, P < 0.001$; comparison type x time: $F_{4,3130} = .7, P = 0.61$; n = 313 cells (Fig. 7*D*); CeN: ANOVA, main effect of comparison type: $F_{1,356} = .3, P = 0.59$; main effect of time: $F_{4,890} = 1.0, P = 0.40$; comparison type x time: $F_{4,1780} = .2, P = 0.96$; n = 178 cells (Fig. 7*F*)]. Thus, neuronal activity during the sucrose CS+ and shock CS+, but not during the sucrose CS− and shock CS+, was more similar than expected by chance.

Activity of Individual Amygdala Neurons Is Correlated with Blood Pressure and Related to the Direction of the Neuronal Response to the Conditioned Stimuli. If neuronal activity in the amygdala contributes to emotional arousal, then measures of arousal should covary with single-cell activity in the amygdala (40–43). To determine whether neuronal activity in the amygdala contributed to the changes in blood pressure we observed during the sucrose CS+ and shock CS+, we looked for relationships between single-cell activity and blood pressure by using multiple linear regression, in which experimental events (such as CS presentations) were used as regressors to account for possible coincident activation of neuronal activity and blood pressure (*SI Materials and Methods* and *Data and Discussion* in the *SI Text*). Remarkably, over 43% of cells (188/434 cells; BLA, 138/312, 44%; CeN, 41/101, 41%) had activity that was correlated with blood pressure, despite using a stringent statistical threshold (α = .0001). In contrast, when the activity data for each cell was shuffled before the regressions, none of the cells' activity was correlated with blood pressure.

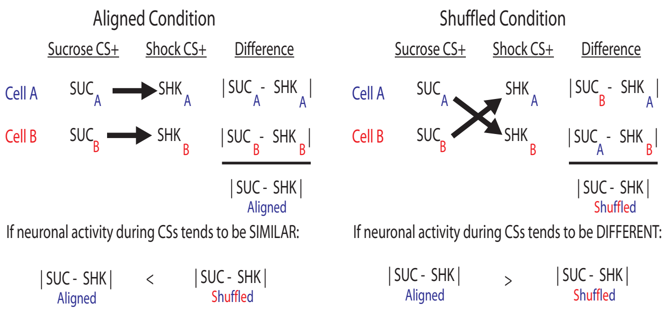

Fig. 6. Design of aligned versus shuffled comparisons. If neuronal activity is similar during the conditioned stimuli, then the difference in the aligned condition should be less than the difference in the shuffled condition. If neuronal activity is different during the conditioned stimuli, then the difference in the aligned condition should be greater than the difference in the shuffled condition.

Consistent with the hypothesis that same cells are important for changes in arousal, the activity of same cells was more likely to be correlated with blood pressure than that of the rest of the population (same, 60/109, 55%; rest, 128/325, 39%; $\chi^2 = 7.5, P < 0.01$). Furthermore, the majority of the directions of the correlations (positive or negative) was congruent with the direction of the neuronal responses (increases or decreases in activity, respectively) to the conditioned stimuli (45/60 cells, 75%), as one would expect if the activity of same cells contributes to increases in blood pressure during the conditioned stimuli. In fact, same cells were more likely to have congruent correlations with blood pressure than selective cells (same: 45/109, 41%; selective: 52/187, 28%; $\chi^2 = 5.1, P < 0.05$) but not more likely to have incongruent correlations with blood pressure (same: 15/109, 14%; selective: 28/187, 15%; $\chi^2 = .01, P = 0.91$).

Discussion

In this study, we demonstrate striking similarities in the activity of amygdala neurons during conditioned appetitive and aversive emotional arousal and strong relationships between neuronal activity in the amygdala and blood pressure—a measure of arousal which is sensitive to lesions of the amygdala in both appetitive (24) and aversive (21, 22) settings. Cells with increases in activity during one CS+ were more likely than would be

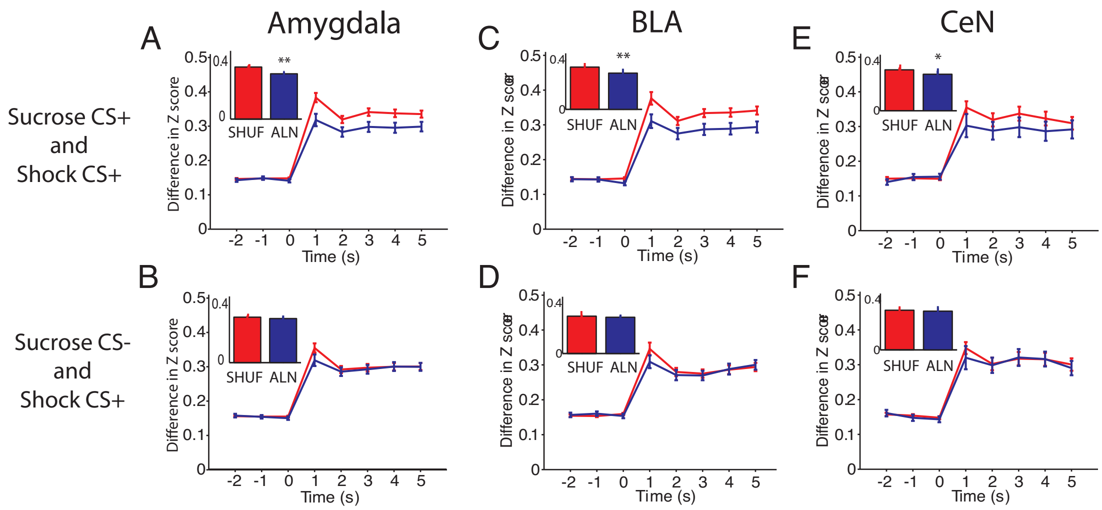

Fig. 7. Comparisons of neuronal activity during the conditioned stimuli across all recorded cells. (*A* and *B*) All amygdala cells (n = 518 cells). (*C* and *D*) BLA cells (n = 313 cells). (*E* and *F*) CeN cells (n = 178 cells). (*A*, *C*, *E*) Comparisons of neuronal activity during the sucrose CS+ and shock CS+. The mean differences in the aligned condition are less than the mean differences in the shuffled condition. (*B*, *D*, *F*) Comparisons of neuronal activity during the sucrose CS− and shock CS+. No significant differences between aligned and shuffled conditions. Large frames and insets as in Fig. 2. * $P < 0.05$, ** $P < 0.0001$.

expected by chance to show the same response to the other oppositely valenced CS+, and the same was true of CS− elicited decreases. Neuronal activity was also more similar than expected by chance during the appetitive and aversive conditioned stimuli when the activity of all recorded cells was considered together. Additionally, the activity of neurons with similar directions of change in firing during conditioned appetitive and aversive stimuli (same cells) was more likely to be correlated with blood pressure than the rest of the population. Same cells were also more likely than selective cells to have correlations consistent with a contribution to conditioned blood-pressure responses (i.e., "congruent" correlations, as described in *Results*). Although other studies report that neurons within the amygdala may contribute to valence-specific encoding (28, 44), our results indicate that an additional substantial proportion of amygdala neurons can contribute to both appetitive and aversive emotional arousal.

Importantly, we found similarities in activity despite the different sensory modalities of the conditioned stimuli, the different outcomes they predicted, and the different behaviors they elicited. We also found that the overall similarity in activity was specific to the comparison between the conditioned appetitive and aversive stimuli, because neuronal activity during the control stimulus and the shock CS+ was not more similar than expected by chance. In fact, the average change in activity during the control stimulus was much smaller than that during the conditioned appetitive and aversive stimuli. Thus, similarities in activity during conditioned appetitive and aversive stimuli are due to the learned significance of the stimuli rather than simple sensory stimulation.

Our results are consistent with single-cell recording studies which noted that some amygdala neurons respond to both appetitive and aversive stimuli (26, 36, 45). However, these studies were either descriptive in nature or were otherwise not designed to quantify the degree of similarity between appetitive and aversive stimuli. Furthermore, our finding that cells with similar responses to conditioned appetitive and aversive stimuli are especially likely to be correlated with blood pressure suggests that these cells play a role in emotional arousal. These results also build on prior studies that found relationships between neuronal activity in the amygdala and measures of arousal but did not test single-cell responses to both appetitive and aversive stimuli (13–17, 40–43).

Although the activity of amygdala cells was more similar than expected by chance when the entire population of recorded cells was considered together, we nevertheless found many cells which were selective for either the sucrose CS+ or shock CS+. This finding is consistent with studies which clearly demonstrated that the activity of amygdala cells is often affected by the associated outcome (27–29) (although selective cells in our study could also be sensitive to the sensory features of the conditioned stimuli). The precise meaning of this outcome-specificity remains to be determined. Outcome-specific cells could be encoding the sensory properties of the anticipated outcome, the positive or negative value (i.e., valence) of the conditioned stimuli or outcomes, or specific affective responses induced by the conditioned stimuli. Because of the different methods of data analysis from those used here, it is also possible that some of the outcome-specific encoding previously reported (27–29) is due to different levels of arousal generated by the anticipated outcomes. For example, cells with increases in activity during both the sucrose CS+ and shock CS+, which we classified as same cells, could be classified as selective for one of the conditioned stimuli by using these alternative methods of data analysis, as long as the magnitudes of the responses to the conditioned stimuli were different. The current study was not designed to explore the properties of selective cells, but this remains an important future goal.

Lesion studies suggest that both the BLA and CeN are important for conditioned increases in blood pressure during fear conditioning (21, 22). Consistent with these studies, we found that the activity of many cells in both regions was correlated with blood pressure. We also found substantial proportions of same cells in both the BLA and CeN, although we note that the likelihood of similar directions of response to the conditioned stimuli was only significant for decreases within the CeN. Given that many cells in the CeN are GABAergic (46), such decreases in activity may disinhibit downstream neurons. Although our data suggest that both the BLA and CeN likely use valence-independent neuronal circuits to produce changes in arousal, they may affect physiological, attentional, behavioral, and mnemonic processes differently via their distinct sets of efferent projections (47). For example, projections from the CeN to the substantia nigra may be important for conditioned orienting responses (48), whereas projections from the BLA through the stria terminalis may be important for modulation of some forms of positively and negatively valenced memories (5). It is also possible that projections from the BLA to the CeN coordinate or permit changes in aspects of emotional arousal (49). The degree to which neuronal activity in the CeN depends on the BLA is an important unresolved issue (50).

Together, our findings indicate that a substantial proportion of neuronal circuits within the amygdala can contribute to both appetitive and aversive emotional arousal. We propose that the function of these neuronal circuits is to prepare the animal to respond to biologically important stimuli, independent of the valence of the stimuli.

Materials and Methods

Ten Long–Evans rats were implanted with blood-pressure transmitters and driveable electrode arrays and, after recovery, water restricted for ≈19 h before training and testing, which proceeded as described in the *Results* section and Fig. 1. Neuronal activity was recorded with commercial hardware and software (Plexon). Single unit isolation and data analysis were performed offline with Offline Sorter (Plexon) and Matlab. At the end of the experiment, current was passed through the recording wires to mark their location. Please see *SI Materials and Methods* for a detailed description of the experimental procedures.

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